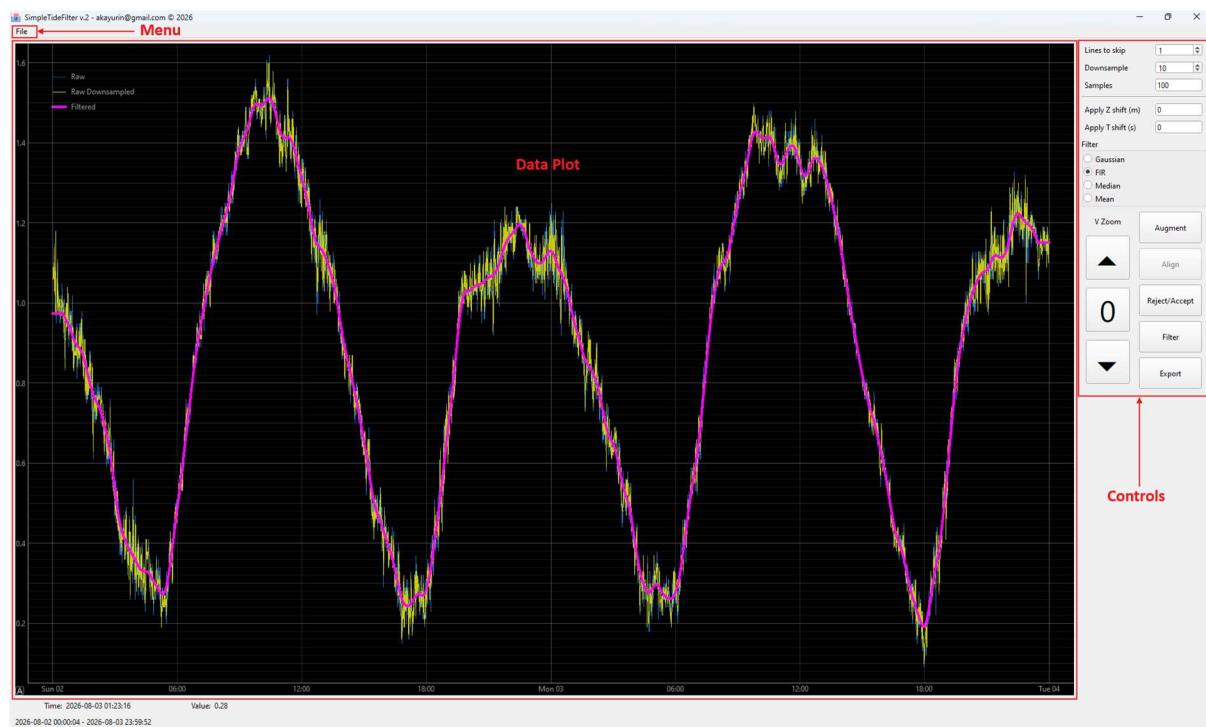


SIMPLE TIDE FILTER MANUAL

SimpleTideFilter application was developed to simplify and speed up noisy water level observations processing before applying in the bathy data processing environment. With SimpleTideFilter it is possible to:

- Decimate observed data;
- Shift tide data vertically and in time;
- Augment and align missing data;
- Reject (de-spike) observed data;
- Apply FIR, Gaussian, Median or Mean filter to smooth decimated data.

The application consists of Main window that includes Menu, Controls and Data Plot.



Source data

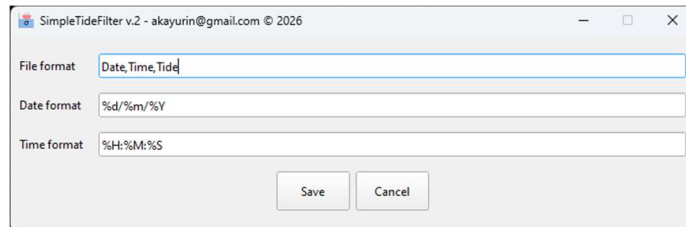
Source water level data should be provided in ASCII format and must necessarily contain Date, Time and Tide in separate fields. The application recognizes space, tab, comma and semicolon as field separators. Note that files may contain other data that will not be considered in the processing. Data format may be configured (see below).

The application is sensitive to foreign data like headers, corrupted strings, etc. and may fail to load data should those exist in the data files. Prior to import data, irrelevant strings should be removed from the files. For example, Qinsy acquisition software adds multiple header lines in the RTK tide files when logging. Those header lines should be removed. If

data file contains single header line, that can be indicated by **‘Lines to skip’** setting in Controls section of the application before loading.

Configuring file and data formats

Application configuration settings are saved in `./_internal/cfg.json` file. Those can be accessed by selecting **‘File -> Edit config’** menu:



Apart from **‘Lines to skip’** setting, there are three settings that may be configured:

- **‘File format’** setting specifies data string format. As noted above, the file string must contain separate Date, Time and Tide fields (regardless their names) separated by space, tab, comma or semicolon. Other fields may also exist the user must tag mandatory fields as ‘Date’, ‘Time’ and ‘Tide’. Other fields may be tagged arbitrarily by unique tags. Field tags must be separated by comma. See examples below:

06/04/2026,00:00:00,-0.19
Date Time Tide

The string above contains only mandatory fields ‘Date’, ‘Time’ and ‘Tide’.

06/04/2026,00:00:00,000,-0.19,ABC,C21
Date Time A Tide B C

The string above contains unnecessary alphanumerical fields that must be tagged arbitrarily. These fields will not be considered in calculations. Another option is to delete the fields from the source file.

- **‘Date format’** setting specifies Date field format. See examples below (for the 6th of April 2026):

06/04/2026 04/06/2026 2026-04-06
%d/%m/%Y %m/%d/%Y %Y-%m-%d

- **‘Time format’** setting specifies Time field format. See examples below:

01:55:55 01:55:55.0 01:55:55.000
%H:%M:%S %H:%M:%S.%f %H:%M:%S.%f

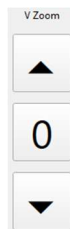
Loading and navigating data

Data may be loaded either from '**File -> Load tide(s)**' or by drag-and-drop files into Data plot. Multiple files loading is supported.

Once loaded, data may be decimated, if necessary, by selecting '**Downsample**' value (every n^{th} sample will be left). Data may be shifted vertically or in time by setting '**Apply Z shift (m)**' and '**Apply T shift (s)**' values.

Note that downsampling and shift values are applied to original loaded dataset. Therefore, all the changes (except augmentation) will be reset. You may need to reject data and/or align dataset again. It is recommended to apply shifts and downsampling before any other manipulations.

Data plot may be zoomed in and out by scrolling **mouse wheel** in the data plot. Vertical and horizontal scale may be adjusted and reset by pressing buttons:



Data rejection/acceptation

Erroneous data or spikes may be rejected and re-accepted. Once '**Reject/Accept**' button is pressed, red rectangle will appear on the Data plot. Press **Left/Right mouse button** to reject/re/accept data. Rectangle may be resized by pressing '**1**'/'**2**' keys. Press the button again to stop rejection/acceptation mode.

NOTE! Rejected data will not be removed from the dataset but interpolated. Consider that when selecting reject region.

Data smoothing

Four filters are available:

- Gaussian;
- Finite Impulse Response (FIR);
- Median;
- Mean.

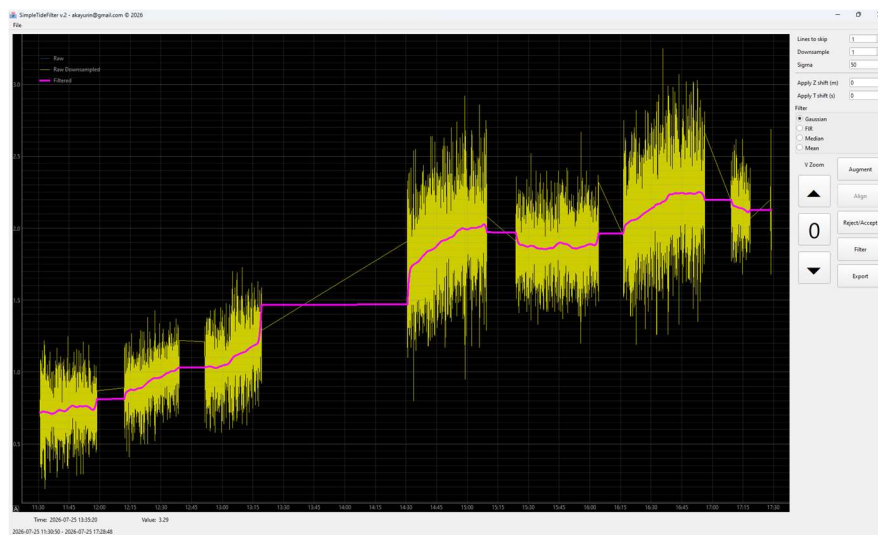
FIR and Gaussian filter are most relevant for smoothing tide data. Filter value may be set by '**Filter value**' control. The control may be named as '**Sigma**', '**Samples**', '**Kernel**' or '**Window**' depending on filter type used. Note that filter value means different

parameters for different filters. Filter takes decimated shifted data. Value should be selected as best fit of the filtered curve. Press '**Filter**' button once '**Filter value**' is set.

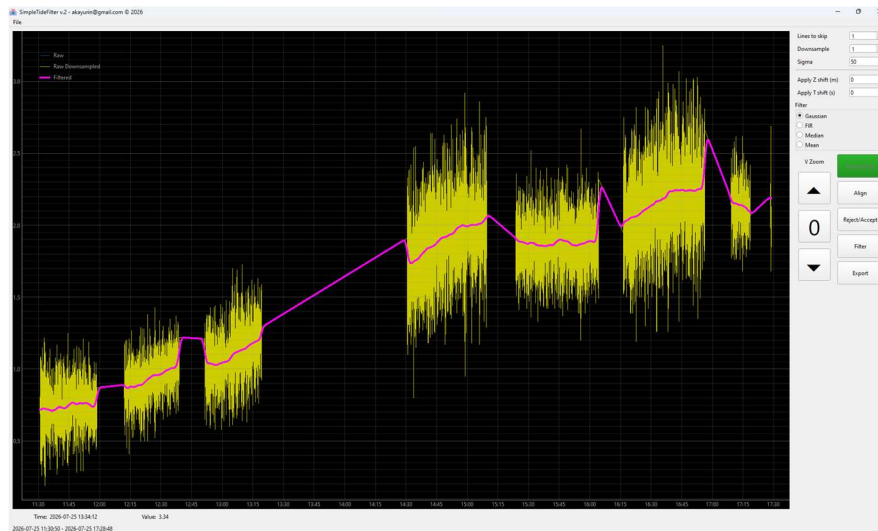
NOTE! Due to filtering algorithms, start and end parts of filtered data become distorted. The size of these parts depends on 'Filter value'. Be cautious when applying in bathy processing. Consider some extra data on both ends of data range.

Data augmentation and alignment

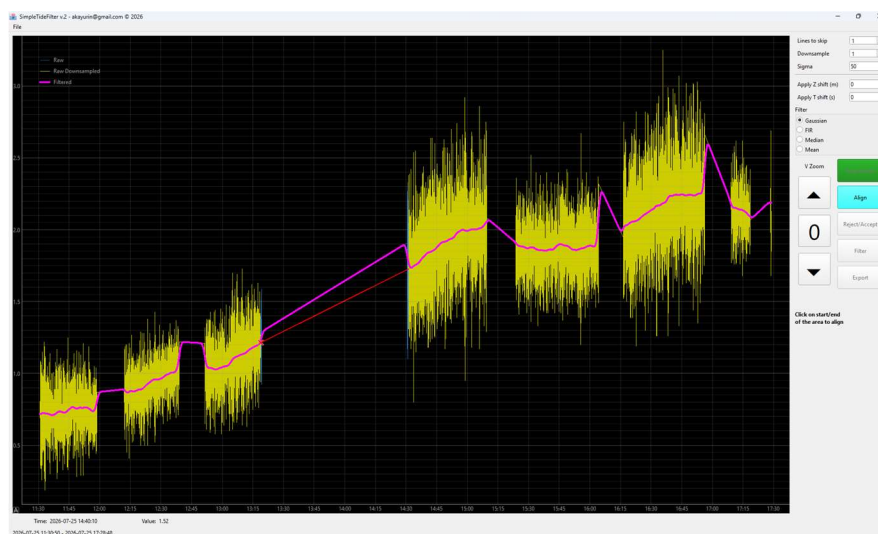
Sometimes processors are provided with water level data acquired only during logging sessions rather than continuous dataset. Depending on variation level, the smoothing result may be incorrect, if logging was started/stopped at the spike time. Ideally, such datasets should be split by logging sessions and filtered separately. Depending on filter strength, this may make data heads and tails unusable.



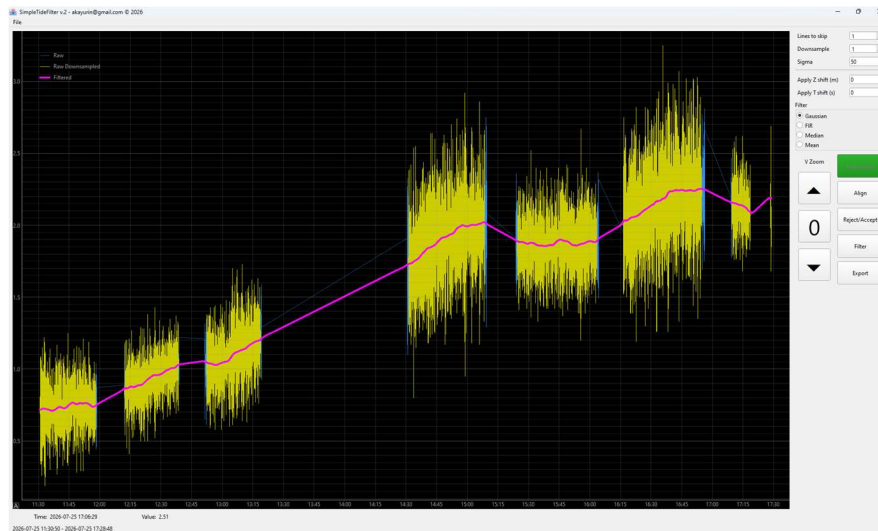
The way to sort it is to augment and align the gaps. Augmentation means adding data points at update rate to missing parts of the dataset. This results in using same filter at the augmented parts. Press '**Augment**' button to augment dataset. Note that it adds data to original loaded dataset and augmentation will be available until dataset is reloaded. Therefore, the augmentation needs to be done only once.



Apparently, tide values at the starts and ends of the missing (augmented) parts are as it was at the start/end of the logging and may not fit general trend of the data due to noise. Aligning data means interpolation of augmented parts between selected points. Those points (end/start) should be selected as close to data trend as possible. To align data, press **'Align'** button and select start/end points. Those will be marked on the screen.



You can align as many segments as possible while the **'Align'** button is pressed. Press it to stop aligning. Then press 'Filter' to apply smoothing to the augmented and aligned dataset.



Result export

Decimated, shifted and filtered result may be exported to ASCII format either from '**File - > Export**' or by pressing '**Export**' button. All data will be saved into single file. 'Date' and 'Time' formats are the same as in the source files. 'Date' and 'Time' fields will be separated by space, 'Tide' field will be separated by comma.